

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EC) Examination****March - Apr 2018****EC241/ EC201.01/ EC201: Digital Electronics & Logic Design****Date: 28.03.2018, Wednesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1.a** Obtain the 1's and 2's complement of the following binary numbers: **[07]**

- | | |
|--------------|-----------|
| i. 1010101 | ii. 10000 |
| iii. 0111000 | iv. 00000 |
| v. 0000001 | |

Q – 2. Answer any two questions. **[14]**

1. Convert the decimal number 250.5 to base 3, base 4, base 7, base 8 and base 16.
2. Reduce the following Boolean expression to four literals.

$$F = (A + C + D)(A + C + D')(A + C' + D)(A + B')$$

3. Perform the Subtraction with the following decimal numbers using 10's complement and 9's complement method.
 - i. 5250-321
 - ii. 753-864

Q - 3 Answer any two questions. **[14]**

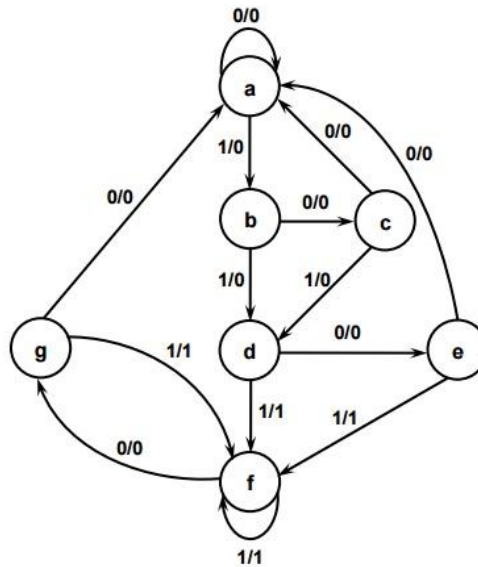
- a. Implement Full Adder combinational circuit with necessary equation and logic diagram.
- b. Simplify the following Boolean function in SOP form:

$$f(w, x, y, z) = \sum (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

- c. Design a combinational circuit to convert BCD to EXCESS-3.
- d. Obtain the simplified expressions in Product of Sums:

$$f(A, B, C, D) = \prod (0, 1, 2, 3, 4, 10, 11)$$

SECTION – II**Q - 4** Reduce the number of state in the following state diagram and draw the reduced state diagram. **[07]**



Q – 5.a Write characteristics and excitation table for SR, J-K, D and T flip-flop. [06]

Q – 5.b Implement the following function with a multiplexer: [05]

$$F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$$

OR

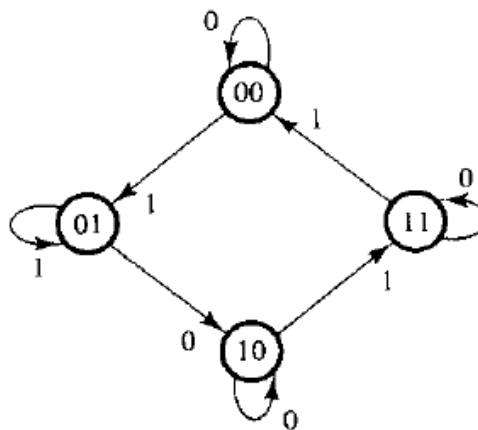
Q – 5.b Design 5x32 Decoder using 3x8 and 2x4 Decoder. [05]

Q – 5.c What is race around condition? Explain the importance of Master slave flip flop. [03]

Q – 6.a Write a short note on Johnson Counter which can generate 8 timing sequence. [06]

Q – 6.b State the differences between Combinational and sequential circuit. [03]

Q – 6.c Design a sequential circuit for the following state table using JK flip-flop. [05]



OR

Q – 6.a Write a short note on 4- bit Universal Shift register. [06]

Q – 6.b Give the difference between synchronous and asynchronous circuit. [03]

Q – 6.c Design a counter that has a binary repeated sequence: 0, 1, 2, 4, 5, 6 using J-K flip flop. [05]